Muhammad Moosa Zafar

Mechanical Design Engineer (EIT) | MEP & HVAC Systems

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WORK EXPERIENCE

Mechanical Design Engineer | AECOM Edmonton, AB *Jan 2025 – Present*

- Sized and selected HVAC equipment — AHUs, unit heaters, exhaust fans, duct heaters, hot water boilers, expansion tanks, and glycol systems — for institutional and commercial projects, ensuring compliance with ASHRAE 62.1/90.1 and NBC.
- Performed ESP calculations for fan selection, pump head, heat loss, infiltration, and thermal expansion stress analysis; conducted psychrometric and HAP analysis to confirm system performance against design-day conditions.
- Produced construction-ready Revit models and drawing packages; resolved MEP clashes through 3D coordination and issued drawing revisions for construction, coordinating across mechanical, electrical, structural, and architectural disciplines.
- Managed CCNs and RFIs across project disciplines, maintaining schedule on fast-track institutional projects and resolving design conflicts promptly.
- Reviewed and approved mechanical shop drawings and submittals for spec compliance; prepared cost estimates and equipment schedules to support project budgeting and procurement.
- Assigned and reviewed work performed by drafting technicians, ensuring deliverables met project standards and construction issue deadlines.

Structures & Inspection Engineer | AECOM Edmonton, AB *Jul 2023 – Dec 2024*

- Inspected steel bridges and overhead sign structures; performed visual weld inspection, ultrasonic thickness testing, and corrosion/section-loss assessment per established codes.
- Developed Excel VBA tools to digitize data collection and automate condition reports for
- Developed Excel VBA tools to digitize data collection and automate condition reports for **350+** overhead sign structures, eliminating **100+** hours of annual manual data entry.
- Prepared technical memos, design reports, and condition assessments under P.Eng. supervision; served as on-site lead for streetscape and bridge rehabilitation works, coordinating daily field operations.
- Assessed fastener integrity, bolt connections, bearings, expansion joints, coating condition, and concrete defects; recommended repairs and further investigation based on deterioration and failure analysis.
- Entered inspection data into bridge management systems to support maintenance planning; trained colleagues on VBA tools developed for inspection and reporting workflows.

ENGINEERING PROJECTS

HVAC ESP Calculator — Desktop Engineering Tool | Independent *2025*

- Developed a Python/Tkinter desktop application to systematically calculate external static pressure losses across multi-section duct systems, automating fitting loss lookups, velocity checks, and total system ESP — replacing manual spreadsheet workflows used in daily design work.
- Packaged as a standalone .exe for distribution to colleagues; incorporated SMACNA fitting loss coefficients and velocity pressure tables to ensure code-compliant outputs.

UT Probe Holder — Magnetic Pipe Scanner | Independent *2026*

- Reverse-engineered and designed a spring-actuated constant-force UT probe holder with magnetic mounting for hands-free pipe scanning (2" NPS to flat); conducted FEA across PLA, PETG, ABS, and PA6-CF to optimize structural performance under spring loading.
- Developed TPU gaskets for probe sealing and vibration dampening; designed modular, field-repairable architecture with interchangeable components to accommodate multiple probe sizes.

EDUCATION

B.Sc. Mechanical Engineering | University of Alberta Edmonton, AB *2023*

TECHNICAL SKILLS

Design & Analysis: HVAC/MEP system design, duct & pipe sizing, ESP, pump head, heat loss, infiltration, HAP analysis, psychrometrics, FEA, ANSYS CFD

Codes & Standards: ASHRAE 62.1/90.1, SMACNA, NBC, NFPA 30/54, CSA B149.1, ASME B31.3, ASME Section VIII

Software: Autodesk Revit, AutoCAD, Inventor, SolidWorks, Fusion 360, Bluebeam, Microsoft Office Suite

Programming: Excel VBA (report automation, Userforms, data tools), Python, MATLAB

CERTIFICATIONS

Registered EIT – APEGA (actively pursuing P.Eng.) | DASSAULT Certified Associate – Mechanical Design | Fall Protection & Confined Space Entry/Monitor Level 2 | WHMIS 2023 | First Aid & CPR/AED Level C